

CU BOULDER / COURSERA

Spacecraft Dynamics Capstone Project

Attitude Dynamics and Control of a Nano-Satellite
Orbiting Mars

Course Report

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Abstract

In this project, the knowledge acquired from the Spacecraft Dynamics and Control specialization is applied to perform attitude control of a nano-satellite orbiting Mars while operating as a daughter-craft to a mother spacecraft in a geostationary Mars orbit (GMO). Through the completion and analysis of eleven project tasks, an attitude control architecture is developed and validated. The resulting simulation enables three operational modes based on the spacecraft mission requirements and relative orbital geometry:

- 1) Science mode: pointing the spacecraft sensor toward the Martian surface,
- 2) Power mode: orienting the solar panels toward the Sun,
- 3) Communication mode: pointing the communication antenna toward the mother spacecraft.

The spacecraft dynamics, reference-frame generation, attitude error evaluation, and closed-loop control simulations are implemented in MATLAB, while mission visualization is performed using AGI Systems Tool Kit (STK).

Chapter 1

Introduction

This project considers a small science satellite orbiting Mars in a Low Mars Orbit (LMO), tasked with collecting scientific data from the Martian surface and relaying the acquired information to a larger mother spacecraft orbiting in a Geostationary Mars Orbit (GMO). In addition to science and communication operations, the spacecraft must periodically orient its solar panels toward the Sun to maintain sufficient onboard power.

Accordingly, the spacecraft mission is divided into three operational modes:

- 1) **Science Mode:** pointing the sensor toward the Martian surface,
- 2) **Power Mode:** orienting the solar panels toward the Sun,
- 3) **Communication Mode:** pointing the communication antenna toward the GMO spacecraft.

Both the LMO and GMO spacecraft are assumed to follow known circular orbits whose orientations are described using Hill reference frames generated through a 3-1-3 Euler-angle sequence.

The LMO spacecraft is equipped with:

- A science sensor aligned with the $+b_1$ body axis,
- Solar panels whose normal vector aligns with the b_3 axis,
- A communication antenna aligned with the $-b_1$ axis.

The objective of this project is to design and validate an attitude determination and control framework capable of autonomously switching between the required operational pointing modes.

Chapter 2

Mission Scenario Definitions

2.1 Mission Orbit Overview

The LMO and GMO spacecraft are both assumed to follow circular orbits around Mars. The orbital motion is modeled using classical two-body dynamics, where Mars is treated as the central gravitational body.

The physical and orbital parameters used throughout the simulations are summarized in Table 2.1.

Table 2.1: Simulation and Spacecraft Parameters

Parameter	Value	Unit
Mars radius R_{Mars}	3396.19	km
Mars gravitational parameter μ_{Mars}	4.28283×10^{13}	m^3/s^2
LMO altitude	400	km
LMO orbital radius r_{LMO}	3796.19	km
GMO orbital radius r_{GMO}	20424.2	km
LMO orbital angular rate $\dot{\theta}_{LMO}$	0.000884797	rad/s
GMO orbital angular rate $\dot{\theta}_{GMO}$	0.0000709003	rad/s

The orbital angular rates are computed using the circular-orbit relation:

$$\dot{\theta} = \sqrt{\frac{\mu_{Mars}}{r^3}}$$

The initial orbit orientation angles of both spacecraft are summarized in Table 2.2.

Table 2.2: Initial Orbit Frame Orientation Angles

Spacecraft	Ω (deg)	i (deg)	$\theta(t_0)$ (deg)
LMO	20	30	60
GMO	0	0	250

Figure 2.1 illustrates the definition of a circular orbit using relative elements along with inertial and Hill frames' base vectors. Note that the orbital elements (Ω, i, θ) can be used as (3-1-3) Euler Angles to obtain Hill frame $\mathcal{H}$ from inertial frame $\mathcal{N}$.

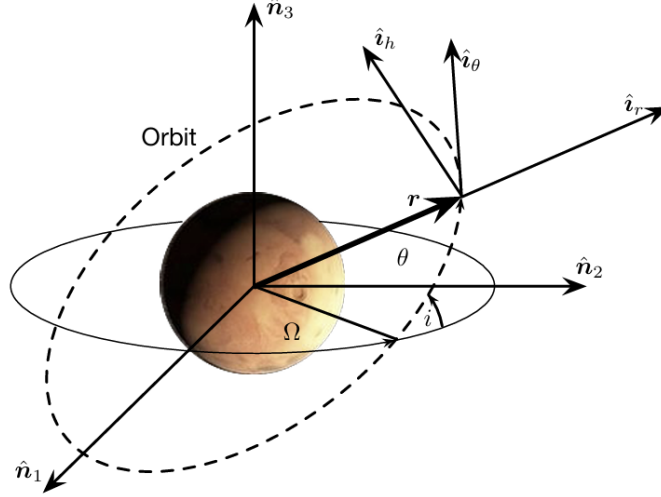


Figure 2.1: Illustration of circular orbital elements alongside inertial frame $\mathcal{N} : \{\hat{n}_1, \hat{n}_2, \hat{n}_3\}$ and Hill frame $\mathcal{H} : \{\hat{i}_r, \hat{i}_\theta, \hat{i}_h\}$ base vectors

2.2 Mission Pointing Scenarios

A body frame as shown in Figure 2.2 is mounted on the spacecraft, assuming to be the principle body frame.

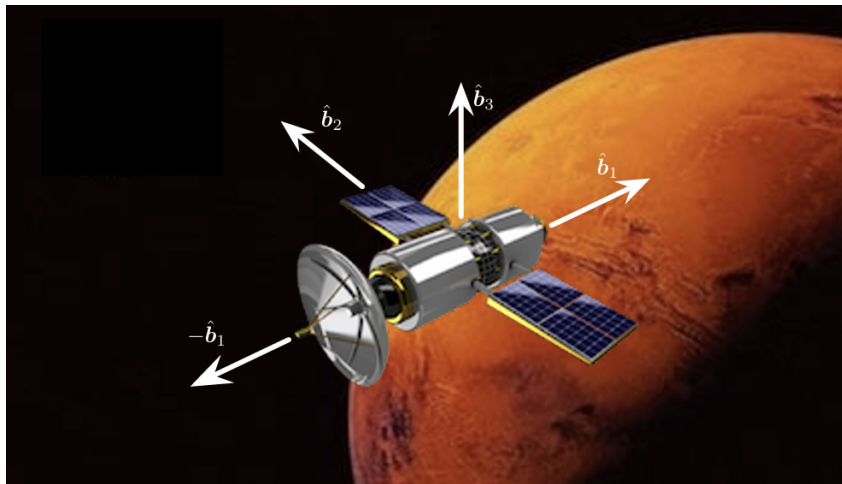


Figure 2.2: Illustration of LMO spacecraft's body frame $\mathcal{B} : \{\hat{b}_1, \hat{b}_2, \hat{b}_3\}$

Three operational spacecraft pointing modes are defined according to mission requirements.

The Sun is assumed to be located at an effectively infinite distance and aligned with the inertial n_2 axis. Therefore, during Sun-pointing operation, the spacecraft solar-panel axis b_3 must align with the inertial n_2 direction.

The spacecraft operational modes are summarized as follows:

- **Power Mode:** active whenever the spacecraft is on the sunlit side of Mars,
- **Science Mode:** active during eclipse when GMO communication is unavailable,
- **Communication Mode:** active whenever the GMO spacecraft lies within the communication visibility constraint.

The communication mode is activated when the angular separation between the LMO and GMO position vectors is less than 35° .

Table 2.3: Mission Pointing Scenario Summary

Orbital Situation	Operational Mode
Spacecraft on sunlit Mars side	Power Mode
Spacecraft in eclipse and GMO visible	Communication Mode
Spacecraft in eclipse and GMO not visible	Science Mode

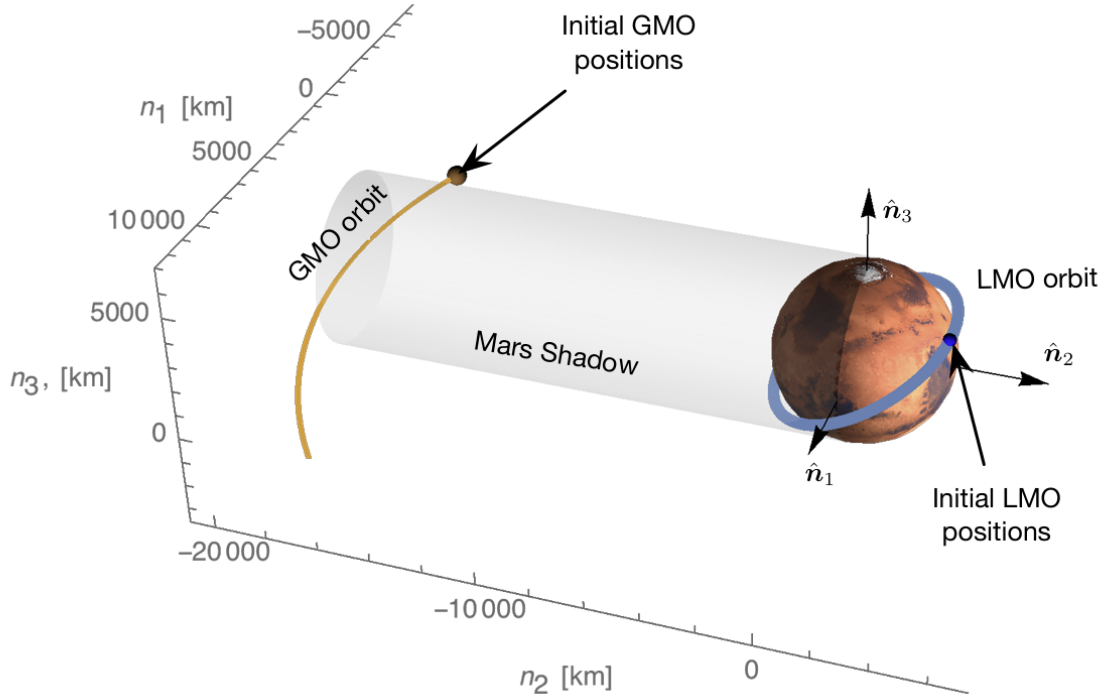


Figure 2.3: Mission Orbit Scenario Illustration

2.3 Spacecraft Attitude States

The spacecraft initial attitude is represented using Modified Rodrigues Parameters (MRPs), while the initial angular velocity is expressed in the spacecraft body frame.

The initial spacecraft attitude is defined as:

$$\boldsymbol{\sigma}_{B/N}(t_0) = \begin{bmatrix} 0.3 \\ -0.4 \\ 0.5 \end{bmatrix}$$

The initial body angular velocity is:

$${}^B\boldsymbol{\omega}_{B/N}(t_0) = \begin{bmatrix} 1.00 \\ 1.75 \\ -2.20 \end{bmatrix} \text{ deg/s}$$

The rigid-body spacecraft inertia tensor is assumed to be:

$${}^B[I] = \begin{bmatrix} 10 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 7.5 \end{bmatrix} \text{ kg}\cdot\text{m}^2$$

The spacecraft is assumed to be capable of generating arbitrary three-axis control torques through an onboard actuator system.

2.4 Attitude Control Law Overview

A proportional-derivative (PD) control law is used throughout this project to regulate spacecraft attitude motion:

$${}^B\mathbf{u} = -K\boldsymbol{\sigma}_{B/R} - P{}^B\boldsymbol{\omega}_{B/R}$$

where:

- $\boldsymbol{\sigma}_{B/R}$ is the attitude tracking error,
- ${}^B\boldsymbol{\omega}_{B/R}$ is the angular velocity tracking error,
- K and P are controller gain matrices.

The objective of the controller is to detumble the spacecraft while driving the body frame toward a desired mission reference frame.

2.5 Implementation Methods

[NEED REVISION BEFORE EVEN BEING WRITTEN]

Chapter 3

Task Descriptions and Solutions

3.1 Orbits

3.1.1 Task 1: Orbit Simulation

Assume the general orbit frame as $\mathcal{O} : \{\hat{i}_r, \hat{i}_\theta, \hat{i}_h\}$. Even though the general orbit frame and Hill frame basically use the same base vectors, their difference is found in their origin. $\mathcal{O}$ frame has its origin aligned with the origin of inertial frame $\mathcal{N}$ – which is the center of Mars. The Hill frame $\mathcal{H}$, however, is mounted on the spacecraft, therefore has the same origin as the body frame $\mathcal{B}$.

Both these frames are obtainable through a (3-1-3) Euler Angle rotation on the inertial frame, though the hill frame origin is also translated to the spacecraft.

Now, using the $\mathcal{O}$ frame, spacecraft position vector is expressed as:

$${}^{\mathcal{O}}\mathbf{r} = r\hat{i}_r$$

Both LMO and GMO orbits are circular, therefore having the same angular velocity $\dot{\theta}$ all over the orbit. Spacecraft velocity vector is expressed as:

$${}^{\mathcal{O}}\dot{\mathbf{r}} = r\dot{\theta}\hat{i}_\theta$$

The MATLAB function `orbitTheta()` takes time, angular velocity and initial angle to calculate the element θ using the following equation:

$$\theta_t = \dot{\theta}t + \theta_0$$

```
1 function theta = orbitTheta(time,thetaDot,theta0)
2
3 theta = rem(thetaDot*time+theta0, 2*pi);
4
5 end
```

Taking the position and velocity vectors from the general orbit frame to the inertial frame is done using the DCM $[NO]$ which can be obtained having all orbital elements at a given time.

$$[ON] = M_3(\theta_t)M_1(i)M_3(\Omega)$$

$$[NO] = [ON]^T$$

$${}^{\mathcal{N}}\mathbf{r} = [NO] {}^{\mathcal{O}}\mathbf{r}$$

$${}^{\mathcal{N}}\dot{\mathbf{r}} = [NO] {}^{\mathcal{O}}\dot{\mathbf{r}}$$

This is achieved by the MATLAB function `orbit_state()` as follows:

```

1 function [r,r_dot] = orbit_state(r,Omega,i,theta,theta_dot)
2
3 % Since the motion is planar and circular, at any time we
   have
4 % r_dot = 0 i_r + r*theta_dot i_theta + 0 i_h
5
6 r_0 = [r;0;0];
7 r_dot_0 = [0;r*theta_dot;0];
8
9 ON = M3(theta)*M1(i)*M3(Omega);
10 NO = transpose(ON);
11
12 % Taking the O frame vectors to the N frame
13 r = (NO*r_0) / 1000;
14 r_dot = (NO*r_dot_0) / 1000;
15
16 % r parameters are expressed in km
17 % r_dot parameters are expressed in km/s
18
19 end

```

3.1.2 Task 2: Orbit Frame Orientation

The Hill-frame basis vectors were constructed directly from the inertial position and velocity vectors:

$${}^{\mathcal{N}}\hat{\mathbf{i}}_r = \frac{\mathbf{r}_{LMO}}{\|\mathbf{r}_{LMO}\|}$$

$${}^{\mathcal{N}}\hat{\mathbf{i}}_h = \frac{\mathbf{r}_{LMO} \times \dot{\mathbf{r}}_{LMO}}{\|\mathbf{r}_{LMO} \times \dot{\mathbf{r}}_{LMO}\|}$$

$${}^{\mathcal{N}}\hat{\mathbf{i}}_\theta = \hat{\mathbf{i}}_h \times \hat{\mathbf{i}}_r$$

To drive a DCM with the base vectors in either frames, we have:

$$[FB] = \begin{bmatrix} {}^{\mathcal{F}}\hat{\mathbf{b}}_1 & {}^{\mathcal{F}}\hat{\mathbf{b}}_2 & {}^{\mathcal{F}}\hat{\mathbf{b}}_3 \end{bmatrix} = \begin{bmatrix} {}^{\mathcal{B}}\hat{\mathbf{f}}_1^T \\ {}^{\mathcal{B}}\hat{\mathbf{f}}_2^T \\ {}^{\mathcal{B}}\hat{\mathbf{f}}_3^T \end{bmatrix}$$

Therefore, to find $[HN]$ we will have:

$$[HN] = \begin{bmatrix} {}^{\mathcal{N}}\hat{\mathbf{i}}_r^T \\ {}^{\mathcal{N}}\hat{\mathbf{i}}_\theta^T \\ {}^{\mathcal{N}}\hat{\mathbf{i}}_h^T \end{bmatrix}$$

A MATLAB function named `orbit_frame_dcm(t)` was implemented to evaluate the time-varying Hill-frame orientation.

```

1 function HN_DCM = orbit_frame_dcm(t)
2 h_dc = 400*1000; % m
3 R_mars = 3396.19*1000; % m
4 r_LMO = R_mars + h_dc; % m
5
6 mio_mars = 42828.3*10^9; % m^3/s^2
7 theta_dot_LMO = sqrt(mio_mars / (r_LMO^3)); % rad/s
8
9 RA_dc = 20 * pi/180; % DC right ascension
10 i_dc = 30 * pi/180; % DC Inclination
11 theta_dc_0 = 60 * pi/180; % DC initial angle
12
13 theta_dc_t = rem(theta_dot_LMO*t + theta_dc_0, 2*pi);
14 [r_dc_t, r_dot_dc_t] = orbit_state(r_LMO, RA_dc, i_dc,
    theta_dc_t, theta_dot_LMO);
15
16 i_r = r_dc_t / norm(r_dc_t);
17 i_h = cross(r_dc_t, r_dot_dc_t) / norm(cross(r_dc_t,
    r_dot_dc_t));
18

```

```

19 i_theta = cross(i_h, i_r);
20
21 HN_DCM = [i_r'; i_theta'; i_h'];
22 end

```

3.2 Reference Frame Orientation

3.2.1 Task 3: Sun-Pointing Reference Frame Orientation

To point the spacecraft solar panels axis $\hat{\mathbf{b}}_3$ at the Sun, the reference frame $\mathcal{R}_s$ must be chosen such that $\hat{\mathbf{r}}_3$ axis points in the Sun direction ($\hat{\mathbf{n}}_2$ in the scenario). To define the frame completely, the first axis $\hat{\mathbf{r}}_1$ is assumed to point in $-\hat{\mathbf{n}}_1$ direction. Therefore, the base vectors will be:

$$\hat{\mathbf{r}}_1 = -\hat{\mathbf{n}}_1$$

$$\hat{\mathbf{r}}_3 = \hat{\mathbf{n}}_2$$

$$\hat{\mathbf{r}}_2 = \hat{\mathbf{r}}_3 \times \hat{\mathbf{r}}_1 = -(\hat{\mathbf{n}}_2 \times \hat{\mathbf{n}}_1) = \hat{\mathbf{n}}_3$$

Similar to the previous task, the analytical expression for the DCM can be obtained using the base vectors:

$$[R_s N] = \begin{bmatrix} \mathcal{N} \hat{\mathbf{r}}_1^T \\ \mathcal{N} \hat{\mathbf{r}}_2^T \\ \mathcal{N} \hat{\mathbf{r}}_3^T \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

The resulting reference-frame DCM can also be obtained using a (3-1-3) Euler Angle of $(\pi, \frac{\pi}{2}, 0)$ rotation on it. MATLAB function `RsN_DCM()` serves this purpose:

```

1 function rotation = RsN_DCM()
2
3 rotation = M1(pi/2)*M3(pi);
4
5 end

```

Since the base vectors of this frame are just a re-arrangement of the inertial frame base vectors, the frame remains fixed relative to the inertial frame, thus:

$$\boldsymbol{\omega}_{R_s/N} = 0$$

3.2.2 Task 4: Nadir-Pointing Reference Frame Orientation

To point the spacecraft sensor platform axis $\hat{\mathbf{b}}_1$ towards the center of Mars (nadir direction), the reference frame $\mathcal{R}_n$ must be chosen such that $\hat{\mathbf{r}}_1$ axis points towards the planet. To completely describe the frame, we assume the second axis $\hat{\mathbf{r}}_2$ points in velocity direction $\hat{\mathbf{i}}_\theta$. Therefore, we'll have:

$$\begin{aligned}\hat{\mathbf{r}}_1 &= -\hat{\mathbf{i}}_r \\ \hat{\mathbf{r}}_2 &= \hat{\mathbf{i}}_\theta \\ \hat{\mathbf{r}}_3 &= \hat{\mathbf{r}}_1 \times \hat{\mathbf{r}}_2 = -\hat{\mathbf{i}}_h\end{aligned}$$

This frame can be obtained using a (3-1-3) Euler Angles rotation on the Hill frame with angles $(\pi, \pi, 0)$.

$$[R_n H] = M_1(\pi)M_3(\pi)$$

We also had previously obtained $[HN]$ in task 2 as a function of time. So,

$$[R_n N](t) = [R_n H][HN](t)$$

In Matlab, this task is done by function `RnN_DCM`.

```
1 function rotation = RnN_DCM(t)
2
3 RnH = M1(pi)*M3(pi);
4 HN = orbit_frame_dcm(t); % Function from task 2
5
6 rotation = RnH*HN;
7 end
```

Since this frame is a re-arrangement of the Hill frame, it also moves with it, thus:

$$\boldsymbol{\omega}_{R_n/N} = \boldsymbol{\omega}_{H/N} = \dot{\theta}\hat{\mathbf{i}}_\theta$$

3.2.3 Task 5: GMO-Pointing Reference Frame Orientation

The communication reference frame was defined such that the spacecraft communication axis continuously points toward the GMO spacecraft.

The relative position vector between the spacecraft was defined as:

$$\boldsymbol{\rho} = \mathbf{r}_{GMO} - \mathbf{r}_{LMO}$$

The communication-frame DCM was constructed from the normalized relative-position vector and orthogonal basis vectors.

Since the reference-frame motion is time varying, the reference angular velocity was obtained numerically using finite-difference differentiation of the DCM.

3.3 Attitude Evaluation and Simulator

3.3.1 Task 6: Attitude Error Evaluation

The spacecraft attitude tracking error was computed relative to a desired reference frame.

The body-to-reference DCM is given by:

$$[C_{BR}] = [C_{BN}][C_{RN}]^T$$

The corresponding MRP attitude tracking error was extracted from the body-to-reference DCM.

The angular velocity tracking error was computed from:

$$\boldsymbol{\omega}_{BR} = \boldsymbol{\omega}_{BN} - [C_{BR}]\boldsymbol{\omega}_{RN}$$

A MATLAB function named `attitudeErrorEval()` was implemented to evaluate both attitude and angular velocity tracking errors.

3.3.2 Task 7: Numerical Attitude Simulator

The spacecraft rotational dynamics were propagated using Euler's rigid-body rotational equation:

$$[I]\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times ([I]\boldsymbol{\omega}) = \mathbf{u}$$

The spacecraft MRP kinematics were propagated according to:

$$\dot{\boldsymbol{\sigma}} = \frac{1}{4}B(\boldsymbol{\sigma})\boldsymbol{\omega}$$

A fixed-step RK4 numerical integrator was implemented to propagate spacecraft rotational motion.

To avoid MRP singularities, shadow-set switching was performed whenever:

$$\|\boldsymbol{\sigma}\| > 1$$

The numerical simulator was validated under both uncontrolled and controlled torque conditions.

3.4 Completing the Mission

3.4.1 Task 8: Sun Pointing Control

A PD attitude controller was implemented to drive the spacecraft toward the Sun-pointing reference frame.

The control torque is defined as:

$$\mathbf{u} = -K\boldsymbol{\sigma}_{BR} - P\boldsymbol{\omega}_{BR}$$

The controller gains were selected using linearized closed-loop dynamics to satisfy desired transient-response requirements.

Simulation results demonstrated stable convergence toward the desired Sun-pointing orientation.

3.4.2 Task 9: Nadir Pointing Control

The nadir-pointing control mode used the same PD control architecture while continuously updating the rotating reference frame.

The controller successfully tracked the time-varying nadir reference orientation while maintaining stable spacecraft rotational dynamics.

3.4.3 Task 10: GMO Pointing Control

The GMO-pointing control mode continuously oriented the spacecraft communication platform toward the GMO spacecraft.

The communication reference frame and corresponding angular velocity were updated dynamically throughout the simulation.

Simulation results demonstrated stable closed-loop tracking of the moving communication reference frame.

3.4.4 Task 11: Mission Scenario Simulation

A complete mission simulation was developed to automatically switch between spacecraft operational modes according to mission conditions.

The mission logic was implemented as follows:

- If the spacecraft was illuminated by the Sun, Sun-pointing mode was activated,
- If the spacecraft was in eclipse and the GMO spacecraft was visible, communication mode was activated,

- Otherwise, nadir-pointing science mode was selected.

At every simulation step:

1. The active reference frame was selected,
2. The reference angular velocity was evaluated,
3. The attitude tracking errors were computed,
4. The PD control torque was generated,
5. The spacecraft rotational dynamics were propagated using RK4 integration.

The complete mission simulation demonstrated stable closed-loop attitude behavior and smooth transitions between operational spacecraft modes.

3.5 Detailed Mathematical Formulation

3.5.1 Orbital Angular Velocity

For a circular orbit, the orbital angular rate is defined as:

$$\dot{\theta} = \sqrt{\frac{\mu}{r^3}}$$

where:

- μ is the gravitational parameter of Mars,
- r is the orbital radius.

The orbital angle evolution is then computed through:

$$\theta(t) = \theta_0 + \dot{\theta}t$$

3.5.2 Modified Rodrigues Parameters

The spacecraft attitude is represented using Modified Rodrigues Parameters (MRPs):

$$\boldsymbol{\sigma} = \hat{e} \tan\left(\frac{\Phi}{4}\right)$$

The corresponding MRP kinematic differential equation becomes:

$$\dot{\boldsymbol{\sigma}} = \frac{1}{4}B(\boldsymbol{\sigma})\boldsymbol{\omega}$$

3.5.3 Shadow Set Transformation

To avoid singularity when:

$$\|\boldsymbol{\sigma}\| > 1$$

The shadow-set transformation is applied:

$$\boldsymbol{\sigma}^S = -\frac{\boldsymbol{\sigma}}{\|\boldsymbol{\sigma}\|^2}$$

3.6 Simulation Results and Discussion

Simulation results demonstrated stable spacecraft attitude behavior under all operational modes. The implemented proportional-derivative controller successfully reduced both attitude and angular velocity tracking errors while maintaining smooth spacecraft rotational motion.

The complete mission simulation validated the autonomous switching architecture between spacecraft operational modes.

3.7 STK Visualization

To validate the MATLAB simulations visually, spacecraft orbit propagation and attitude motion were reconstructed using AGI Systems Tool Kit (STK).

The generated visualization confirmed:

1. Correct orbital motion of both spacecraft,
2. Proper nadir and Sun-pointing orientations,
3. Communication alignment during GMO visibility windows,
4. Stable attitude transitions throughout mission operation.

3.8 Appendix: MATLAB Function Overview

Table 3.1: Main MATLAB Functions Used in the Project

Function Name	Description
<code>orbit_state()</code>	Computes spacecraft inertial position and velocity vectors
<code>orbitTheta()</code>	Computes orbital angle evolution
<code>orbit_frame_dcm()</code>	Generates Hill-frame DCM
<code>RsN_DCM()</code>	Returns Sun-pointing reference-frame DCM
<code>RnN_DCM()</code>	Returns nadir-pointing reference-frame DCM
<code>RcN_DCM()</code>	Returns GMO-pointing reference-frame DCM
<code>attitudeErrorEval()</code>	Computes tracking errors
<code>simulate_mode()</code>	Simulates spacecraft attitude dynamics
<code>getReferenceFrame()</code>	Performs automatic mission-mode selection

Conclusion

This project presented the complete development and implementation of an attitude dynamics and control framework for a nano-satellite orbiting Mars. Multiple mission pointing modes were developed, including Sun-pointing, nadir-pointing, and GMO communication-pointing configurations.

The spacecraft dynamics, reference-frame generation, attitude tracking evaluation, and closed-loop control architecture were successfully implemented in MATLAB and validated through numerical simulation.

The final integrated mission simulation demonstrated stable spacecraft attitude behavior while autonomously switching between operational mission modes according to orbital conditions.

Future improvements may include higher-fidelity environmental disturbance modeling, actuator limitations, sensor-noise modeling, and advanced nonlinear control strategies.